

NITHILAN KUMARAN

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Education

University of Wisconsin-Madison

Aug. 2024 – May 2028

B.S. in Computer Engineering, Minor in Applications of AI in Engineering

Madison, Wisconsin

- **Relevant Coursework:** Data Structures (Java), Object-Oriented Programming (Java), Algorithms, Machine Organization (C), Computer Graphics, Game Design I, Human and Computer Interaction

Experience

Neureal

May 2026 – Present

Software Engineering Intern

San Mateo, California

- Profiled and optimized runtime performance on a live multiplayer Roblox title, reducing latency by 5% across gameplay systems.
- Developed state-driven enemy AI in Luau on a shared combat framework, implementing windup, fire, and recovery attack phases synchronized across multiplayer clients, shipped to live gameplay.
- Built the Go daemon behind NeuBridge, an internal MCP developer tool bridging Roblox Studio to VSCode, implementing live console log streaming and DataModel snapshot services that power editor autocomplete and path diagnostics.

Software Engineering Intern – Summers 2023 & 2024

- Designed and shipped core in-game shop and transaction systems in Lua for a live Roblox title with **164M+ lifetime visits**, implementing search algorithms and end-to-end purchase flow.
- Ported a React-based UI architecture to Roblox Lua, translating component-based design patterns, state management, and event-driven logic into a game engine compatible implementation.
- Implemented dynamic and lazy loading for component-based Roblox UI, cutting load work for off-screen content to enable smooth scrolling and responsive content delivery in large catalogs.

Sentari AI

Sep 2025 – Dec 2025

Software Engineering Intern

New York City, NY

- Reduced manual publishing overhead to near zero by architecting a multi-stage content automation pipeline integrating n8n, LLM APIs, and WordPress, deployed to production through Vercel.
- Hardened pipeline reliability with scheduling, retry logic, and structured error handling, improving deployment consistency and reducing operator intervention.

Carrier Global Corporation

May 2025 – Aug 2025

Software Systems Intern

Atlanta, Georgia

- Integrated Amazon Bedrock and Amazon Q into Carrier's SAP environment, automating payment and billing workflows across distributed enterprise systems.
- Developed an AI-driven financial prediction service backed by SQL databases and vector stores, writing queries and data models to surface insights for senior engineering stakeholders.
- Led SAP UAT for Spot Buy Purchase Orders and Purchase Requisitions, validating end-to-end workflows across distributed systems and reducing production defects by 15%.

Projects

Meridian | *In-Memory Cache Server, C++20, CMake*

Jan 2026 – Jul 2026

- Built a Redis-style cache server from scratch in C++20 with an O(1) LRU cache, kqueue event loop, and consistent hash sharding.
- Scaled in-process cache throughput 12x (838K to 10.3M ops/sec) with lock-stripped concurrency, verified race-free via ThreadSanitizer.
- Sustained 165K req/sec end-to-end at p99 under 1 ms across 100 concurrent TCP connections.
- Trained a logistic-regression eviction policy (sklearn) with hand-written C++ inference, beating LRU hit rate 1.4x.

SageRec | *C++17, pybind11, CMake, Python*

May 2026 – Present

- Built a C++ neighbor sampler with a CSR graph backend for GraphSAGE mini-batch training on MovieLens, running 111x faster than a matching Python reference implementation.
- Trained GraphSAGE end to end on the native sampler, beating a matrix factorization baseline on ML-100K (Recall@10 0.048 vs 0.038).

Open Source

Laya (Convai) | *Open-Source Decision Engine, External Contributor, 23K+ GitHub Stars, Python*

Sep 2026

- Merged 6 PRs as an external contributor: an opt-in embedding shortlist for high-cardinality choice sets, and language detection across every string in nested mapping/dict state.

Technical Skills

Languages: C++20, Go, Python, Java, JavaScript/TypeScript, Lua/Luau, SQL, Bash

Systems & Cloud: Concurrency, Multithreading, MCP, REST APIs, PostgreSQL, Docker, Git, CI/CD, AWS (Bedrock, Amazon Q)

Frameworks & Engines: React, Next.js, Node.js, CMake, Roblox Studio

Certifications: AWS Certified Cloud Practitioner, AWS Certified AI Practitioner

Writing: 36 engineering articles on Medium/Stackademic, 10,000+ total reads (medium.com/@Sathariels)